C. H. Adams Data

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1 Overview

- Observer & Site: C. H. Adams, UK (likely near London).
- Period Covered: August 1819 – May 1823.
- Content: Nearly daily full-disk sunspot drawings with accompanying tabular counts of sunspot groups and individual spots.
- Observation Method: Used an asterisk notation distinguishing direct view ("*") from reflection view (no asterisk); observation times frequently noted.
- Significance: Provides dense daily coverage—including many spotless days—during an under-documented period, improving the calibration and low-activity baseline of ISN V3.

1.1 Structure & Content

- Tabular layer: Per-day entries of number of groups (NG), number of spots (NS), spotless indicator, observation time, method flag, and comments. Wolf numbers $W = 10 \cdot NG + NS$ can be derived.

- Image layer: 338 scanned manuscript drawings requiring orientation and dewarping.
- Metadata layer: Observer, site, instrument, method, provenance (Mittheilungen vol./page, folio IDs), and database foreign-key links (observer/rubric).

1.2 Coverage & Volume

- Drawings: 338
- Total dated entries: 1,056 $\rightarrow$ 308 spot-days + 748 spotless days
- Comparison: Zürich Mittheilungen (402 entries) cover only 94 spotless days; Adams adds ~650 new spotless statements—critical for establishing quiet-Sun baselines.

1.3 Scientific Importance

- Completeness: Captures numerous explicit zero-spot (spotless) days often absent elsewhere.
- Method sensitivity: Direct vs reflection views allow analysis of observational bias and variance.
- Future use: Once oriented, drawings will provide positional and areal data for hemispheric sunspot numbers and dynamo studies.
- Historical value: Offers a rare, continuous dataset bridging early 19th-century solar cycles for long-term variability research.

1.4 FARSuN Progress

- Retrieval: Led by Pavai with Lefèvre & Ritter; extracted tables (sunspot_metadata-ADAMS_...) aligned with Mittheilungen references.
- Quality Control: Removed ambiguities in NG/NS; planned dispersion checks between Adams/Mittheilungen/Pavai sets; validated with Gruithuisen overlap (Aug–Sep 1819).
- Digitization: Orientation and dewarping (e.g., folio 15) via ScanTailor Advanced; limb registration standardized; reflection   direct transformations tested (R. Arlt).
- Metadata encoding: Observer, site, method, time, spotless flag, provenance linked to the historical database.
- Accessibility (planned): Release through EPN-TAP/VO-TAP at ROB with QC flags, versioning, and query fields (method, spotless, provenance).

1.5 Integration into ISN V3 — Near-Term Steps

- Finalize coverage calendar and authoritative day list.
- Publish day-level comparison log (Adams vs Mittheilungen/Pavai).
- Model observational biases and uncertainties by method.
- Test inclusion of Adams as an independent node in ISN V3 stitching to assess baseline impact.
- Release v1 (NG/NS + spotless + method/time + provenance); v2 to include positional/areal data.

2 Gaps & Follow-ups

- Check for material beyond 1819–1823.
- Verify completeness of direct/reflection flags.
- Report orientation accuracy metrics before releasing coordinates.
- Confirm rights, assign DOI, and register in IVOA.
- Explore citizen-science validation and future scientific publications.

In essence: The C. H. Adams sunspot archive offers an exceptional, method-rich dataset that strengthens the early-19th-century solar record. FARSuN's ongoing digitization and calibration work will make it a cornerstone reference for the International Sunspot Number V3 and broader studies of solar variability and climate linkage.